Weixuan Yuan

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Research interests: LLM post-training with verifiable feedback (formal specifications, executable checks); agent harnesses for mathematical and algorithmic research; formal verification in Lean 4.

Available for internships from January 2027 (3–6 months, location flexible).

Education

ETH Zürich — Ph.D. in Computer Science May 2025–Present
Advisor: Prof. Rasmus Kyng. Research Assistant in the same group, October 2024–April 2025.

ETH Zürich — M.Sc. in Data Science September 2022–October 2024
Willi Studer Prize (2025), awarded to the best graduate of the M.Sc. in Data Science program.

Xi'an Jiaotong University — B.Sc. in Mathematics September 2018–July 2022
Honors Youth Program

Selected Research

AuXSpec: Learning to Autoformalize Program Specifications via Executable Postconditions

Co-first author · under review at ICLR 2027. Training LLMs to translate natural-language programming problems into formal specifications in Lean.

- Proposed *executable postconditions*: an executable counterpart of each declarative Lean specification that can be checked on any test case via `#guard`, turning specification correctness into a cheap automatic signal for data filtering and a core component of the RL reward.
- Co-designed the Lean specification format and co-built a multistage LLM pipeline for specification synthesis, verification, and repair, producing 3,037 formalized programming problems and ~19.5K SFT samples.
- Reviewed 300 benchmark problems and audited 60 model-generated specifications.
- With SFT + RL, a 27B model improved SpecGen pass@4 from 0.7% to 60.3% on AuXSpec-Test and from 21.9% to 84.7% on Verina.

Deterministic Gomory–Hu Trees — FOCS 2025

Co-authored the first deterministic almost-linear-time algorithm for Gomory–Hu trees (all-pairs minimum cuts), running in $m^{1+o(1)}$ time on graphs with m edges. Presented the paper with Wuwei Yuan at FOCS 2025. [slides]

Publications

AuXSpec: Learning to Autoformalize Program Specifications via Executable Postconditions
Co-first author. Under review at ICLR 2027; manuscript not yet publicly available.

Deterministic Almost-Linear-Time Gomory-Hu Trees
Amir Abboud, Rasmus Kyng, Jason Li, Debmalya Panigrahi, Maximilian Probst Gutenberg, Thatchaphol Saranurak, **Weixuan Yuan**, Wuwei Yuan.
IEEE Symposium on Foundations of Computer Science (FOCS 2025). Authors listed alphabetically. [paper]

Statistics (undergraduate work, before 2023; † equal contribution)
Xuehu Zhu†, **Weixuan Yuan**†, Zongben Xu, Wenlin Dai. Optimal subsampling for principal component analysis. *Science China Mathematics* 68, 2993–3016, 2025. [paper]
Xinru Jia†, **Weixuan Yuan**†, Xingqiu Zhao, Xuehu Zhu. Optimal subsampling for estimation of dimension reduction directions. *Scandinavian Journal of Statistics* 53(1), 575–611, 2026. [paper]

Technical Skills

Programming: Python, Lean 4.
Theory: Algorithms; probability and statistics; optimization, including gradient-based methods.